THE FUNCTOR OF FLATNESS IN DIMENSIONS AT LEAST N IS AN FPQC SHEAF AND IS LIMIT PRESERVING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a chart reduction identifying the functor with a quasi-affine chart functor, applied to base changed fpqc coverings for the sheaf property and to a manufactured filtered limit for limit preservation

Editorial note. The label lemma-flat-dimension-n names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The situation defining the functor classifying the points where a module is flat in dimensions at least n, and the chart lemma reducing the functor to a quasi-affine situation over the base
- The sheaf property for the fpqc topology and limit preservation in that quasi-affine situation
- The behaviour of fpqc coverings and of filtered limits of affine schemes under base change

Source reference. Original lemma 05MV: sources/stacks-chow/tags.txt, source lines 7284–7284. Prerequisite pointer invoked at proof line(s) 2139; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CWF: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2030–2030. Prerequisite pointer invoked at proof line(s) 2136; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05MT: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2085–2085. Prerequisite pointer invoked at proof line(s) 2123; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 05MT.*

(1) *The functor F_n satisfies the sheaf property for the fpqc topology.*
(2) *If f is quasi-compact and locally of finite presentation and $\mathcal{F}$ is of finite presentation, then the functor F_n is limit preserving.*

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Proof. Proof of (1). Suppose that $\{T_i \to T\}$ is an fpqc covering of a scheme T over Y. We have to show that if $F_n(T_i)$ is nonempty for all i, then $F_n(T)$ is nonempty. Choose a diagram as in part (1) of Lemma 0CWF. Denote F'_n the corresponding functor for $\varphi^*\mathcal{F}$ and the morphism $U \to V$. By More on Flatness, Lemma 05MV we have the sheaf property for F'_n. Thus we get the sheaf property for F_n because for $T \to Y$ we have $F_n(T) = F'_n(V \times_Y T)$ by Lemma 0CWF and because $\{V \times_Y T_i \to V \times_Y T\}$ is an fpqc covering.

Proof of (2). Suppose that $T = \lim_{i \in I} T_i$ is a filtered limit of affine schemes T_i over Y and assume that $F_n(T)$ is nonempty. We have to show that $F_n(T_i)$ is nonempty for some i. Choose a diagram as in part (1) of Lemma 0CWF. Fix $i \in I$ and choose an affine open $W_i \subset V \times_Y T_i$ mapping surjectively onto T_i. For $i' \geq i$ let $W_{i'}$ be the inverse image of W_i in $V \times_Y T_{i'}$ and let $W \subset V \times_Y T$ be the inverse image of W_i. Then $W = \lim_{i' \geq i} W_i$ is a filtered limit of affine schemes over V. By Lemma 0CWF again it suffices to show that $F'_n(W_{i'})$ is nonempty for some $i' \geq i$. But we know that $F'_n(W)$ is nonempty because of our assumption that $F_n(T) = F'_n(V \times_Y T)$ is nonempty. Thus we can apply More on Flatness, Lemma 05MV to conclude. $\square$